

Input: initial minimum x_0 ; maximum saddle index $k_{\max}$

Output: reaction route map graph G

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1 Function Explore(parent, xstart, Vasc):
2   c ← MHiSD(xstart, Vasc)
3   if G contains a node c' ≈ c then
4     | add edge (parent–c') to G                                     // rediscovered
5   else
6     | add node c to G
7     | if index(c) < kmax then mark c upward-pending
8     | else mark c upward-done
9     | if index(c) > 0 then mark c downward-pending
10    | else mark c downward-done
11    | add edge (parent–c) to G
12  end
13 end

14 Function Upward(n):
15   // n has index k; projected-Hessian eigenpairs ( $\lambda_1, v_1$ ),... ascending
16   choose vp: eigenvector of the smallest eigenvalue  $\lambda_p > 0$                                      // p = k+1
17   Vasc ← {v1, ..., vk, vp}                                     // parent's k modes + new one
18   foreach s ∈ {+ $\varepsilon$ , – $\varepsilon$ } do
19     | Explore(n, xn + s vp, Vasc)
20   end

21 Function Downward(n):
22   // n has index k > 0; spawns 2k runs in total
23   for i ← 1 to k do
24     | Vasc ← {v1, ..., vk} \ {vi}                                     // the other k–1 unstable modes
25     | foreach s ∈ {+ $\varepsilon$ , – $\varepsilon$ } do
26       | Explore(n, xn + s vi, Vasc)
27     | end
28   end

29 Function Orchestrator(x0, kmax):
30   G ← graph with one node n0 = CriticalPoint(x0) of index 0
31   mark n0 upward-pending; mark n0 downward-done                                     // a minimum cannot descend
32   repeat
33     | while ∃ upward-pending node n with index(n) < kmax do
34       | Upward(n)
35       | mark n upward-done
36     | end
37     | while ∃ downward-pending node n with index(n) > 0 do
38       | Downward(n)
39       | mark n downward-done
40     | end
41   until no upward-pending node has index < kmax and no downward-pending node has index > 0
42   return G
43 end

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Algorithm 1: Reaction route map construction via high-index saddle dynamics. *Notation.* x is a molecular geometry, with x_0 the input local minimum and x_n the geometry of node n ; G is the reaction route map graph (nodes are critical points, edges are their connections); $k_{\max}$ is the highest saddle index searched and $\text{index}(\cdot)$ is the Morse index (number of negative Hessian eigenvalues). Each critical point stores the eigenpairs (λ_i, v_i) of its projected Hessian (rigid-body modes removed), ordered by ascending eigenvalue: v_i is an eigenvector and λ_i is its eigenvalue, with $\lambda_i < 0$ an unstable (ascent) mode. n_0 is the root node (the input minimum as an index-0 point); n is the node currently being expanded, c the critical point returned by a run (a candidate node) and c' an existing node matched to it; *parent* is the node the run was launched from, and the edge linking it to c records their connection. V_{asc} is the ascent subspace—the modes that MHiSD maximises (it minimises on the orthogonal complement). In UPWARD, $p = k + 1$ is the rank of the perturbation mode (the smallest positive eigenvalue). ε is a small fixed step size and $s \in \{+\varepsilon, -\varepsilon\}$ the signed perturbation, so $x_n + s v$ displaces the geometry by ε along $\pm v$. Finally, $\approx$ tests structural equivalence up to rigid-body motion and a tolerance.